

Dhwani Shingala

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Scala, Java, SQL, C/C++

Frameworks: NextJS, NodeJs, FastAPI, LangGraph, Flask, ReactJS, Hadoop, Spark, Kafka, CrewAI

Cloud: AWS (EC2, S3, RDS, Bedrock, EMR, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions)

Databases: PostgreSQL, MongoDB, MySQL, CosmosDB, ChromaDB, Supabase

Tools: Claude Code, Cursor, Copilot, Bash, Git, Linux, Grafana, Airflow

EXPERIENCE

Software Engineer I

June 2024 – Present

UI Health

Chicago, IL

- Architected **Lumen**, a multi-agentic therapy platform in **LangGraph**, designing a supervisor-to-specialist routing system with parallel sub-graph execution, reducing task latency by 40% and enabling real-time therapy sessions for 200+ users.
- Built a real-time speech-to-agent ingestion pipeline using **Deepgram Nova-3** enhanced with **FAISS**-based **RAG retrieval** over a clinical knowledge base, cutting retrieval latency by 55%.
- Engineered a **FHIR**-compliant healthcare integration layer with **REST API** tooling exposing appointment scheduling and automatic flagging, backed by a **two-pass guardrail pipeline (Amazon Bedrock Guardrails)** that reduced harmful outputs below 0.1% in production.
- Designed a **HIPAA-compliant backend infrastructure** on AWS with PostgreSQL, S3 audit logging, along with a JWT-authenticated API gateway with rate limiting, sustaining **99.9% uptime** under peak load.
- Composed an **Airflow**-orchestrated **Spark** ETL pipeline integrating the **Fitbit API** to ingest and process time-series biometric data from 500+ participants daily, serializing outputs to **Parquet** on **S3** and reducing processing time by 60%, efficient physiological feature extraction and trend analysis.
- Implemented a **FastAPI**-based backend handling 1000+ daily requests coupled with a **MongoDB** database to streamline structured data ingestion for the ACE-D trial, reducing manual entry latency by 75%.
- Built a real-time research metrics dashboard in **NextJS/Typescript** aggregating data across 5+ concurrent health trials, cutting report generation from 4 hours to under 15 minutes and enabling same-day operational decisions.
- Designed and shipped an end-to-end **NIH** safety reporting platform in **NextJS/Typescript** with automated **SQL**-driven data pipelines, eliminating 20+ hours of manual quarterly work, achieving 99.9% data accuracy, maintaining DSMB regulatory compliance.
- Engineered **REST API** and **SQL** ETL pipelines to parse, validate, and normalize 100K+ unstructured dietary records into a structured relational schema, reducing data preparation time by 80%.

Backend Engineer

Sept 2022 – Apr 2023

Institute for Plasma Research Center

Gandhinagar, India

- Revamped legacy Flask backend modules, re-architecting access control and CMS module integrations for Joomla 4.2 to **modernize authentication workflows** and improve user dashboard management efficiency.
- Optimized PostgreSQL query performance across core CMS data models, **reducing redundant joins** and introducing indexing strategies to improve system response times by 20% and reduce average query execution time by 35%.

PROJECTS

SalesOps – Multi-Agent Orchestration | Python, LangGraph, ChromaDB

May 2025 – July 2025

- Orchestrated a modular multi-agent system (LangGraph + FastAPI + React) **automating B2B sales workflows** end-to-end, with role-based agents like lead research, outreach writing, and email drafting.
- Architected a Hybrid RAG system (ChromaDB, TinyDB) for vector storage, memory and retrieval, optimizing cost and latency by **dynamically routing queries** between local (Ollama) and API-based (OpenAI) LLMs.

Distributed Algo | Scala, Akka, Grafana, Prometheus

Oct 2024 – Dec 2024

- Implemented 10+ distributed algorithms in **Scala** using the **Akka** actor framework, spanning leader election and routing protocols, with support for message-passing paradigms.
- Engineered an automated simulation engine driven by JSON-based plans, incorporating weight-throwing termination detection and end-to-end observability via **Lightbend Telemetry**, **Prometheus**, and **Grafana**.

EDUCATION

University Of Illinois Chicago

Chicago, IL

Master of Science in Computer Science

Aug 2023 – May 2025

Dharamsinh Desai University

Gujarat, India

Bachelor of Technology in Information Technology

Aug 2019 – May 2023